JORC Code Reporting

PROFESSIONAL CERTIFICATE

MODULE 1 VIDEO TRANSCRIPTS

Why the JORC Code?

Why the JORC Code? A good question to start the course with. Why was the JORC Code written and what was the imperative?

The idea that a professional code of practice was needed to ensure minimum standards of public reporting had been talked about for a while. The imperative came in 1969 on the back of a high-profile share trading fiasco. At the time, nickel was in demand due to the Vietnam War, and nickel supply was low out of Canada, their main supplier. Interest in new nickel discoveries was high and when Poseidon NL, a small West Australian company announced the discovery of nickel at a place called Mount Windarra investors reacted to cause one of the largest share booms the Australian Securities Exchange, or ASX, had seen.

On October 1, 1969, the Poseidon directors issued an historic report to this Adelaide Stock Exchange

before the start of trading. The statement read, "Further to the report of the recovery of nickel and copper sulfides on September 29, the directors of Poseidon NL announce that the assays received to date of the first completed drill holes of PH2 at Windarra, WA, were 40 feet of 3.56 nickel and 0.55% copper." The grades were based on a drill hole that included massive sulfides.

The consulting geologists, Burrill and Associates Pty Ltd went on to say, "The mineralised zone has an indicated length of 1000 feet and a minimum width of 65 feet."

As a direct result of these preliminary results in interpretations, Poseidon's share value rose from around a dollar, eventually reaching $280 in five months. Poseidon's share price bonanza inspired would-be investors into looking for other affordable mining stocks. Numerous new companies were listed on the ASX, some of which didn't even have mining leases, let alone a viable project.

Reality struck in March 1970, when the share price crashed along with many other nickel hopefuls, most with less tangible prospects than Poseidon and fortunes were lost.

As a footnote. Windarra was eventually mined by Western Mining, although it was a small deposit.

So, what happened?

At the time, in September 1969, of the reporting, Poseidon really only had early-stage assays estimated to the nearest percent. The actual assays ended up being nearly 11% lower than originally reported, but the public were not given this information until five years later.

The eventual recovered nickel mine grade was about half that reported pre-mining, and that, in conjunction with the falling nickel prices, resulted in the mine closing in 1976. Poseidon NL was de-listed from the stock exchange and many investors lost their money. There was a loss of confidence in the ASX resulting in share prices plummeting across the mining industry.

The other significant issue was that the consulting geologist, Burrill, was also a substantial shareholder in Poseidon, putting into question the independence of the public report.

In an effort to turn confidence around, the Australian federal government initiated the Rae Commission, which resulted in the then AMIC, or AMIC, and AusIMM, forming the Joint Ore Reserves Committee in 1971. Its purpose was to

provide a mandatory system for the public reporting of exploration results, mineral resources, and ore reserves according to the levels of confidence in geological knowledge and technical and economic considerations.

Today, JORC is a committee formed from members of the AusIMM, the now MCA, and the AIG, with representatives from the Australian Securities Exchange, the ASX, and the Financial Services Institute of Australasia, FINSIA, and also an observer from the Association of Mining and Exploration Companies, AMEC, and the accounting profession. The JORC committee is also on the Committee for Mineral Reserves International Reporting Standards, CRIRSCO, and partnered with the International Council on Mining and Metals.

The code has been propagated in many countries around the world as the preeminent code from which public reporting is made.

As a footnote, the Poseidon boom and bust experience also led to the formation of the national corporate regulator, now ASIC, the Australian Securities and Investment Commission.

Interview with Steve Hunt

Rene Sterk interviews Steve Hunt, Chairman of the Australasian Joint Ore Reserves Committee (JORC).

Rene To shine a bit more light on the purpose of the JORC Code, I'm speaking with Steve Hunt, who is the current Chairman of the JORC Committee in 2018.

Steve, can you just explain why you're such a strong proponent of the Code, and can you perhaps go into why it is important to have a set of rules?

Steve Yeah, thanks, Rene. I always emphasise that the JORC Code is there primarily for investors, and so investors can understand our industry better.

It's really important that we have a clear and common language, and definitions when we're reporting in the mining industry. And so, the commonality of the definitions, the commonality of the reporting framework and classifications is really important regardless of whether we're at back at the exploration end, whether we're in a development stage or whether we're in an operating mine.

The core definitions have to have a meaning that's well-established and well understood, and it's really a way of making sure that the understanding sticks, and that then people can invest on that basis.

This gives investors confidence, and I think it's really important that they can have a clear understanding. I would ask the people watching this is, would you invest in something if you really didn't understand what the terminology meant?

Without these features, I think it would be really quite difficult for the mining investment to operate appropriately, and certainly, neither investors or regulators, or shareholders would be happy without these sorts of frameworks.

Rene Thanks, Steve, that's really clear. So, I guess it's important for investors to have an appropriate level of assurance of the quality of the information on which they base their investment decisions. Do you think that the codes will keep evolving?

Steve I think the Code has always evolved, and will continue to keep evolving. The evolution of the Code has been a feature all the way through. Probably one of the largest steps we had was with the JORC 2012 update in which the 'if not, why not' concept and the Table 1 enhanced reporting were introduced.

And I think that was an important evolutionary stage to get a better balance between the transparency elements, and I suppose the competence and materiality elements.

It's where that plays, is that it means that people are being asked to report about the stage of the work right across the spectrum, and it makes it clearer to investors which parts are perhaps very complete to a very good level, which parts are still developing, and to get a balance between the risk and the opportunity.

And again, it just emphasises that you should be able to pick up a Public Report, and make sense out of it from an investor's perspective.

In terms of where the evolution of the Code is going, JORC has been a foundation member of CRIRSCO, we've been part of the international framework for a very long time, and the commonality of the global reporting codes, I think comes back to all of them having adopted the JORC key principles approach.

So, you could say the JORC DNA is in place right across the world. The understanding by the investors in the market is very strong, and that's why JORC is sort of, is accepted, and is in use in markets outside of Australasia, as well as, I suppose, the high interest in local countries developing their own CRIRSCO template JORC-type codes for their own environments.

So yeah, look, the evolution will still happen, but I think it's the proof of why it's important is there, and that we've still continued to develop the footprint globally over the last 20 or 30 years.

Rene Thanks, Steve. And in terms of key take home messages, is there anything you can share with the participants in order to get maximum value from this course?

Steve I think it's to engage fully. In terms of the course, you'll find that this stretch is probably out of your area of comfort. There are a lot of people on the course will perhaps primarily work in the exploration end, some might work in operating mines, some might be high frequency reporting users of the Code.

We've designed the course really to take you out of your comfort zone a bit.

It's just as important for an exploration geologist to understand the downstream needs of the mining engineer as it is for the mining engineer to understand what the exploration process is about.

And of course, everyone needs to understand what the ASX and ASIC reporting frameworks and rules are about as well.

Rene Thanks for your contribution, Steve.

Public Reporting

A Public Report is a report that is prepared in accordance with the JORC Code for the purposes of informing investors and potential investors and their advisors. Any public reporting of Exploration Results, Mineral Resources or Ore Reserves must be based on and fairly reflect internal documentation that has been prepared by a Competent Person in accordance with the JORC Code.

A Competent Person is a mineral industry professional who is a member of, or a fellow of the AusIMM or the Australian Institute of Geoscientists, the AIG or of a recognised professional organisation or RPO.

Listed companies are obliged to inform the investor and the public of information significant to that company. Information that may have an impact on its share price. The companies have a range of reporting obligations including annual and quarterly reports which will be discussed later as it is an ASX listing requirement not a JORC Code requirement. The reporting, however, must comply with the JORC Code.

Let's get clear on three important terms used in the JORC Code.

A **significant project** is an exploration or mineral development project, which is significant to the market value or operations of the company. If the project is prominent in the announcement, it means that it's significant to the company. The ASX Listing Rules use the term 'Material Mining Projects' to refer to something similar to the JORC definition of significant project, but with a slight variation. We will discuss this further when we cover the ASX Listing Rules.

Material change is that which could be a change in the estimated tonnage or grade or in the classification of the Mineral Resources or, Ore Reserves. Whether there has been a material change in relation to a significant project must be considered by taking into account all of the relevant circumstances, including the style of mineralisation.

The term **'if not, why not'** was introduced to the Code in 2012. It means that each item listed in the relevant section of Table 1 must be discussed and if it is not discussed then the Competent Person must explain why it has been omitted from the documentation.

Please take note of these definitions and the JORC Code reference to these terms as we will use these terms often, and they will form part of the course assessment.

To comply with both the JORC Code for significant projects and the ASX Listing Rules for material mining projects for reporting initial results or estimates or for reporting results or estimates which have materially changed from when they were last reported, the following information needs to be prepared.

The Competent Person's internal documentation is always required. The Public Report addressing all material information in the body of the report, including those points identified as requiring prominent reporting in ASX Listing Rules 5.7, 5.8, and 5.9. And as an appendix to the Public Report, a completed Table 1 for the relevant sections of the table and as identified in the ASX Listing Rules on an 'if not, why not' basis.

Let's go through that again. The JORC Code reporting can be applied to internal documentation, but it must always be applied to public reporting. And a completed Table 1 must be an appendix to the public report for significant projects when reporting initial results or estimates or reporting results or estimates which have materially changed from when they were last reported.

Public Reporting Example: CuDeco

We've been talking about the JORC Code and ASX Listing Rules requirements for public reporting. Let's now look at an example of an entity that didn't follow the JORC Code and ASX Listing Rules and what occurred as a result.

On 29th of June 2006, CuDeco, which was then known as Australian Mining Investments, announced an interim mineral resource. The Public Report stated an Inferred Resource with no further detail. That's right. What you see here is all that was announced about the Mineral Resource.

A week later, on July 5th, the ASX slapped a trading halt on CuDeco and subsequently a suspension pending further releases of information.

A week after that, on July 14, a subsequent release was issued saying CuDeco was working on a response. By this time, CuDeco had been suspended from trading for nine days.

Three days later, on July 17, CuDeco lodged a revised announcement, outlined the basis for the original Mineral Resource estimate. Following the lodgement, the company was reinstated to ASX listing.

The revised announcement stated the basis on which they had developed the resource, that it was effectively an Inferred Resource with a length, a depth, a width, a density, and a calculation of how deep it is and whether there is extrapolations to it. You may have noticed that CuDeco actually changed it from 2.04% copper equivalent to include 1.57% copper with additional cobalt and gold. They included the additional elements as they believed they were recoverable. An average grade of assays for 21 holes assayed was used.

You may suggest CuDeco's estimation method was rudimentary. However, poor estimation methods are not a breach of the JORC Code. The code is not designed to dictate how to estimate, only how to report it.

As a consequence of CuDeco breaching the JORC Code, it not only lost nine days of trading, but it experienced a loss of reputation impacting the company, shareholders, and the industry as a whole. It wasn't until 2010, more than three years later, that an appropriate Mineral Resource was reported by the company.

If CuDeco had been required to include a Table 1 in its Public Report, the substandard approach to their estimation methodology would've been transparent to shareholders. Table 1 was introduced into the Code in 2012.

The Joint Ore Reserves Committee (JORC) and the JORC Code

The Joint Ore Reserves Committee was established in 1971. Its role is to produce and review the JORC Code. The committee comprises 15 selected volunteers made up of representatives from the groups shown on the slide.

After a series of reports released by the committee in the 1970s and 80s.The first code was released in 1989. It was included in the ASX Listing Rules as the mandatory reporting code, and was enforced in law by the Australian Securities Commission.

In other words, a Competent Person's responsibility to write Public Reports to the requirements of the code is backed by a legal requirement. This aspect of the JORC Code has been a key feature adopted by similar international standards. Indeed, many countries have adopted the JORC Code itself.

The JORC Code is the standard for companies reporting to the ASX, the New Zealand Stock Exchange, and the Port Moresby Exchange. It is a mandatory reporting standard for members is regarded as best practice by the Minerals Council of Australia and the Financial Services Institute of Australia. It is accepted in Canada, Johannesburg, London, Singapore, and Hong Kong.

So what does the code do? It stipulates a minimum standard of reporting, Exploration Results, Mineral Resources, and Ore Reserves. It provides extensive guidelines on the criteria to be considered when preparing reports, and it describes a mandatory system for the classification of tonnage or grade estimates according to geological confidence and technical or economic considerations.

The code is not prescriptive. For example, it does not regulate the procedures a competent person must use to estimate and classify the Mineral Resources or Ore Reserves, and it does not regulate a company's internal classification or reporting systems.

Individuals may report public report non-compliance through a complaint mechanism managed by the regulatory bodies such as the AusIMM or the AIG, and through the ASIC and ASX for Public Reports. Breach investigations are not managed by JORC. Company breaches are dealt with by the ASX and breaches of Competent Persons are dealt with by the professional organisations in which they are a member, such as the AusIMM.

The AusIMM and the AIG have a Code of Ethics and charters that are responsible for maintaining the appropriate professional development standards that people are required to undertake to maintain their Competent Person's status. If a complaint is made, the association's committees judge whether the case against a Competent Person should be upheld. If the Competent Person is judged to have reported or behaved inappropriately, the professional organisation can ban the person from future membership, which impacts their ability to be a Competent Person.

Overview of the JORC Code

The current JORC Code as at September 2022 is the 2012 Edition. The Code is currently in revision.

A key aspect of the JORC Code is classification. It provides a mandatory system for the classification of tonnage or grade estimates according to geological confidence and technical or economic considerations. The classification system enables the Competent Person to communicate their confidence in the estimates to the users of the Public Report. The users, such as investors, are able to compare the reports of mineral assets because of the standardisation of reporting that the JORC Code has enabled.

The JORC Code is central to reporting confidence in the public arena, but we need to remember a number of factors will determine what is actually written in the reports. Inputs include company management, finance, government, investment, the community, and the estimators themselves.

Figure 1 from the JORC Code articulates the general relationship between Exploration Results, Mineral Resources, and Ore Reserves, and the relative confidence in the information for each classification. Only these classifications can be used in Public Reports.

At the lowest level of confidence in relation to potential Mineral Resources is Exploration Results. Exploration Results include drill results, geophysical results, geochemical results, and anything else reported that relates to exploration and is not associated with Mineral Resource estimates.

Mineral Resources can be estimated on the basis of geoscientific information and some input from other disciplines. Classifications include Inferred, Indicated, and Measured. These classifications reflect the increasing geological knowledge and confidence in continuity.

Ore Reserves, which are a modified subset of the Indicated and Measured componentsof the Mineral Resources as shown in the dashed outline, require Modifying Factors affecting the extraction to be considered. Ore Reserves are estimated with inputs from a range of disciplines.

Modifying Factors are considerations used to convert Mineral Resources to Ore Reserves. These include but are not restricted to mining, processing, metallurgical, infrastructure, economic, marketing, legal, environmental, social, and government factors.

In other words, regardless of the quality of the data, these are estimates. The Ore Reserve estimates are guided by the Modifying Factors to generate either Probable or Proved Ore Reserves depending on the inputs from the geological model, but also the relative confidence of the various Modifying Factors. Confidence in the mining method or the metallurgical recovery or the geotechnical or hydrological information can heavily impact the final Ore Reserve reported.

Confidence in the mining method or the metallurgical recoveries will be covered in later sections.

So, in summary, the JORC Code is a code for reporting, not estimating. It's about how to report, not how to do the work behind the scenes. It is up to the lenders and investors to assure themselves of the quality of the estimate.

Principles of the JORC Code

There are three principles governing the operation and application of the JORC Code: Transparency, Materiality, and Competence. Transparency and Materiality are the guiding principles that a Competent Person must consider and address when preparing a Public Report. Competence refers to the expectation surrounding the Competent Person themselves.

Let's take a closer look.

Transparency is information that must be provided in such a way that it is clear and unambiguous and that a reader can comprehend the data and not be misled by the information or omission of data.

For example, reporting a combined Indicated and Inferred Mineral Resource as a total in the headline but not providing the individual classification figures until page eight, where the vast majority of the proportion is Inferred is not transparent reporting.

Importantly, Public Reports should be written free of jargon and written as simply as possible.

The Materiality principle states that all relevant information that investors and their professional advisors would reasonably require and reasonably expect to find in order to make a reasoned and balanced judgement that must be in the report. The Competent Person must provide explanatory commentary on the material assumptions underlying the declaration of Exploration Results, Mineral Resources, and Ore Reserves. If the information is not included, the Competent Person must explain why not.

For example, failing to mention that in the maiden mineral resource of a project that the gold is highly refractory and does not provide all the material information that an investor requires.

Of course, Materiality for a large mining company compared to a junior exploration company would expect it to be quite different.

Lastly, the person doing the estimating and reporting must be competent and take responsibility for their own work. They must be suitably qualified, experienced, and must be a member of a professional body that enforces a code of ethics.

Table 1 of the JORC Code is a checklist of assessment and reporting criteria to be considered by the Competent Person in preparing the Public Report. For significant projects, Table 1 is required when reporting Exploration Results, Mineral Resources, and Ore Reserves for the first time, or if there are material changes from the last Public Report.

The Competent Person must comment on the relevant sections of Table 1 and explain if not, why not if they have nothing to report. Reporting on an 'if not, why not' basis ensures that it is clear to an investor whether items have been considered and deemed of low consequence, or that they have not yet been addressed or resolved at all.

Competent Persons, Responsibility and Liability

Who is deemed a Competent Person, one suitably qualified to write a Public Report?

The JORC Code defines a Competent Person as a minerals industry professional who is a Member or Fellow of the AusIMM, the AIG, or of a recognised professional organisation. These are listed on the JORC and ASX websites. These organisations have enforceable disciplinary processes, including the powers to suspend or expel a member.

A Competent Person must have a minimum of five years of relevant experience in the style of mineralisation or type of deposit under consideration and in the activity in which the person is undertaking.

If the Competent Person is preparing documentation on Exploration Results, the relevant experience must be in exploration.

If the competent person is estimating or supervising the estimation of Mineral Resources, the relevant experience must be in the estimation, assessment, and evaluation of Mineral Resources.

If the Competent Person is estimating or supervising the estimation of Ore Reserves, the relevant experience must be in the estimation, assessment, evaluation, and economic extraction of Ore Reserves.

This definition of a Competent Person conforms with the standard definition of the Committee for Mineral Reserves International Reporting Standards or CRIRSCO.

Let's look at this definition a little closely.

Appropriate professional membership. This requirement is clear. At least five years relevant experience. The key qualifier here is 'relevant'. Relevant means it is not always necessary for a person to have five years experience in each and every type of deposit in order to act as a Competent Person. They must have relevant experience in the style of mineralisation or type of deposit being reported upon.

It is not necessarily sufficient that someone has spent 40 years in a mine. What is important is that they understand the process of estimation sufficiently enough to understand where errors may stem from.

A good litmus test is for the aspiring Competent Person to ask themselves, "Could I face my peers and demonstrate my experience to a group of peers?" If in doubt, the person should seek professional guidance from appropriately experienced colleagues or decline to act as a Competent Person.

As the reporting estimator, the Competent Person has significant responsibility and liability. To this end, the Competent Person should seek ways to protect themselves from scrutiny. The JORC Code plays a role here. If a Competent Person follows the Code with honesty and diligence, this will provide a strong measure of due diligence protection.

The Competent Person has various statutory or common law liabilities that can occur if they fail to apply to appropriate care and diligence. Legal liability is complex and dependent on many specific legal conditions. However, in significant cases, imprisonment is a possibility as well as liability for financial damages claims. In Australia, if a Competent Person is proven guilty of making a false or misleading statement or has omitted relevant information, then a maximum penalty of $22,000, or five years imprisonment, or both may be issued.

Through its Ethics Committee, the professional organisation in which a non-compliant Competent Person is a member may elect to warn, reprimand, suspend, or expel a Competent Person. Expulsion means they can no longer be a Competent Person.

The "AusIMM Monograph 30 Guide to Good Practice" provides some valuable information for the Competent Person.

Okay, some take home points.

The JORC Code provides the regulation of the estimator and not the estimation method. As the estimator, the Competent Person has accountability through their self-regulating professional body.

We can all ignore things that make us cringe, but to improve the industry, we all have the responsibility to take a stand and report any issues observed.

The Competent Person and Clause 9, Competence and Responsibility

In all Public Reports, a Competent Person must include a written consent statement as shown in this example, and also available as a template on the JORC website.

Under the ASX Listing Rules of 5.6, 5.22 and 5.24 the JORC Code pertaining to Competence and Responsibility, the statement must include the following information:

- the name of the Competent Person,
- the name of the Competent Person's firm or employer,
- the name of the professional organisation, of which they are a member, and their member number,
- that they understand the requirements of the JORC Code,
- that they have five years of relevant experience,
- they must disclose any potential conflict of interest such as shares in the company to ensure transparency.

If they do not have a potential conflict, they must state this also.

The Competent Person must state the Public Report is based on documentation compiled by themselves and agree to the form and context of the report. In other words, the style in which the Public Report has been written is also the responsibility of the Competent Person.

It is common for Competent Persons to have shares under an employee share purchase plan. This must be disclosed but is unlikely to be judged material.

Review an example of such a declaration here.

If a subsequent Public Report contains information that is being re-reported from a previous report and has no new information, then it does not need to be re-signed by a Competent Person. However, the original report must be referenced and a statement made by the company that it is not aware of any new information or data that materially impacts the result that was given in the original report. In the case of estimates of Mineral Resources or Reserves, the company must confirm that all material assumptions and technical parameters underpinning the estimates in the relevant Public Report continue to apply and have not materially changed. Having said this, all annual reporting of Ore Reserves must be signed by a Competent Person, even if there is no new information or data as described in the JORC Code under Reporting General.

The JORC Code Competence and Responsibility clause is subtly different to the ASX Listing Rule 5.23, which deals with the same matter in a similar manner. However, under the ASX Listing Rule 5.23 the information must have been reported in accordance with the rule 5.7, 5.8, or 5.9 of the new disclosure rules for the exemption from the Competent Person's sign off for subsequent reports to be available. That is, you cannot repeat information last reported under the 2004 JORC Code without a new Competent Person sign off on the Public Report.

In certain limited circumstances during acquisitions and takeovers, as long as the results are not adopted by the reporting company you may repeat the previous company's results reported under a pre-2012 JORC Code.

Interview with Jacqui Coombes

Rene Sterk, interviews Jacqui Coombes, Chair and Interim CEO of Mining 3 (formerly Managing Director & CEO, Amira Global and Member of the Joint Ore Reserves Committee).

Rene As a fundamental principle of the JORC Code Competency is one of the central pillars.

To discuss the importance of this concept and its implications further, I am talking with Jacqui Coombes.

Jacqui completed her PhD in competency within the framework of the JORC Code, which revealed important insights into the functioning of the Code.

Jacqui, can you share some of your key findings with us and show us how this relates to the JORC Code? What do people need to be aware of when we talk about competency?

Jacqui Well, Rene, I think one of the most important things to be aware of is that the current definition of Competent Persons is based on our understanding from the 1970s.

In reality, the expectation of the market and expectation of investors extends well beyond what we have documented in the Code.

Competency itself extends well beyond just knowledge. There's a layer there of having skills, having an ability to reason, having an ability to understand the risks that have been presented by other technical experts, and to be able to synthesise that into an overall reasoned risk for the investor.

To be able to do that takes a lot more than five years' experience in a particular field in the industry. And what I found from the analysis of competency in the assessment of the reasoning, as it relates to the JORC Code items, was that a Competent Person needs to have at least 10 years industry experience, five of which are in the area that they're reporting to.

That's Exploration Results, Mineral Resources, Ore Reserves. And that in order to gain a level of expertise that provides them with the ability to reason across the JORC Code, they need to have done at least in the Mineral Resources space at least 15 resource estimates across two different settings to enable them to assess the nuances that are particular to a project, and to have done at least five reconciliation studies.

Now, that sounds like a lot of work, but if you think about an analogy would be downhill mountain biking, for example. We might have experienced trudging along the city streets on our bicycles.

That doesn't necessarily equip us with a proficiency to engage in downhill mountain biking. We do need to have a level of applied experience that can build our understanding of the nuances of the context.

Rene And there are significant consequences for Competent Persons for not getting it right, aren't there? You've been involved or have been on the sidelines of some high profile cases. Can you comment on what people need to be aware of in terms of liability?

Jacqui When a Competent Person is signing off their consent to a public release they are entering into the realm of the legal sphere.

They are entering into the sphere of the market, and shareholders take their shares fairly seriously. So, they are relying on the Competent Person to provide, I suppose, the due diligence and understanding that they're not able to do on each and every project.

So there is a responsibility by the Competent Person to provide an objective analysis of the information, a thorough understanding of the risks, and then to provide their authority or their authorisation of the information that's going to the market.
Indeed, the public disclosure is the responsibility of the board of a company, but that public disclosure cannot get through to the market unless there is a Competent Person's signed consent.

And in providing that consent, you're essentially taking on the liability that you as a Competent Person are authorising the public disclosure and that there's nothing materially wrong with the project or the disclosure, and an important clause in the JORC Code is that the Competent Person will not remain silent on any aspect that is material to the disclosure.

So, you are providing a level of assurance to the investor that the public release is in accordance with your understanding.

Rene Right. That's some really useful things for people to consider. And is there anything else with regards to the JORC Code from a competence point of view that you want to share with us?

Jacqui I'd suggest one of the most important aspects of being a Competent Person is having sufficient experience, exposure, and understanding of the mine value chain.

Being part of a community of technical experts in different fields and being able to understand the reasoning that the others bring to your view and to your perspective, and indeed to the communication.

Rene Thanks, Jacqui.

International Codes, Guidelines and RPOs

In this video, we'll summarise some other professional bodies and guidelines that have a relationship with the JORC Code.

Let's start with CRIRSCO. The Committee for Mineral Reserves International Reporting Standards is a group of representatives or organisations that are responsible for developing principles-based mineral reporting codes and guidelines.

Take a look at the list of CRIRSCO members as of 2022 or visit the CRIRSCO website.

CRIRSCO publishes and maintains a free-of-charge international reporting template, much of it drawn from the JORC Code 2012, which countries can leverage to develop their own reporting codes. In this way, a level of consistency of reporting standards for market-related reporting and financial investments is being established across the world. The most recent CRIRSCO template edition was released in November 2019.

The CRIRSCO template is also the basis for other standards in the extractive industries.

There are some guidelines that have a relationship with the JORC Code. For example, the Australian Guidelines for the Estimation and Classification of Coal Resources, which was updated in 2014 and aligns with the JORC Code. Although in reality, the JORC Code is what is used for reporting coal resources.

The coal classification guidelines, however describe some terminology common to the coal industry.

The Guidelines for Reporting Resource and Reserve Estimation for Brines is another associated guideline, important for reporting things like lithium brines.

Registered professional organisations or RPOs are those professional organisations other than the AusIMM and AIG as having been approved by the ASX as accredited organisations to which Competent Persons must belong for the purpose of reporting Public Reports for Exploration Results, Mineral Resources and Ore Reserves.

The JORC website has a current list of these RPOs grouped into UK and European, USA, South African, Canadian and the Chilean, Russian groups.

To become an RPO, an organisation must meet specific requirements such as self-regulation, members must have an accredited academic qualification and an enforceable code of ethics, a disciplinary power and a report when requested on their ethical procedures to JORC.

Interview with Ian Goddard

Rene Sterk interviews Ian Goddard, Past Chairman of CRIRSCO and former member of Executive CRIRSCO and JORC.

Rene The JORC Code is, of course, not the only adopted reporting code in Australia by the ASX, but has also been adopted by the STX in Singapore, the NZX in New Zealand, the HKEX in China, and under an international system of mutual recognition reports produced in accordance with the JORC Code may also be recognised and suitable for use in countries which use other minerals reporting standards.

Here to explain some of the nuances in the appropriate use of the JORC Code outside Australia is Ian Godard, JORC Committee member and past chair of the Committee for Mineral Reserves International Reporting Standards, CRIRSCO.

Ian, can you tell us a little bit about the international use of the JORC Code and about any differences that perhaps RPO members reporting under the Code should be aware of?

Ian Yeah, thanks Rene.

The JORC Code is internationally respected.

It was the first of the codes and it's maintained its leadership in code development over the years through revisions.

CRIRSCO has a template and that's used by countries that intend to become members to develop their own codes.

Initially, it was largely based on the JORC Code, where currently in the final stages of a 2018 revision, and incorporated into that revision are a number of aspects of the JORC 2012 code.

The changes that have happened there, just as some of the other features of other codes are incorporated.

So JORC and its representatives on CRIRSCO have played a major role in developing and maintaining the CRIRSCO template.

JORC and the Canadian National Instrument are the codes or reporting standards that are most widely used throughout the world.

As you said, the JORC Code is the code for Australian, New Zealand and Port Moresby. It can also be used in Hong Kong and Singapore stock exchanges.

As well as that, there are four other areas which have accepted the JORC Code as one of the codes to be used and that's Europe, Brazil, Canada, and Kazakhstan.

Now, those last four also have their own code but they accept the JORC Code for reporting.

Of course, any Competent Person who reports against the JORC Code and those other jurisdictions must be conscious that there may be other regulatory requirements there in their reports. You mentioned RPOs, the recognised professional organisations.

This is a system where Competent Persons from other countries may report against the JORC Code.

There's a list of such recognised professional organisations on the ASX and the JORC websites.

Any Competent Person who meets the requirements of their professional organisation that is recognised has to also be aware that there are some differences between their home country codes and the JORC Code.

Some of them aren't major. Like in JORC we use Ore Reserves and all the others it's Mineral Reserves, but that's not the main issue.

Where they have to be careful be conscious of the fact that in JORC, for example we have the 'if not, why not' provisions.

And in some cases it's mandatory to consider all aspects of Table 1. And that can be for the the first report of a significant project for a company or whether there's a material change in the resources or the reserves for that company.

They must be very conscious that they give the details that are required in all the various elements on Table 1. And also with JORC they have to be conscious of the fact that forward-looking statements like production targets and exploration targets may cause greater scrutiny from the regulators.

Rene Right, and I'm interested.

Are the consequences of failing to report in accordance with the JORC Code the same, what I mean by that is what happens if, for instance, an AusIMM member is included as a Competent Person in a Public Report to the AIM in London where the report includes several mentions of adherence to the JORC but has not addressed most of the issues required on the Table 1?

Ian Well, the process within Australia is quite well known. Both professional societies, the AusIMM and the AIG have a similar process.

So complaints in Australia against JORC are investigated by the professional societies. I've been on the Ethics Committee of the AusIMM for many years now, so I'm aware of the types of alleged breaches that are referred to it, the investigation and the outcomes from that investigation.

But both professional societies require you not only to abide with the Code, but to abide with the laws of the country that you're operating in.

So if a Competent Person reports against the JORC Code dealing with a property in Indonesia say, they have to abide by the Code but also be conscious of any laws there.
I can't comment on AIM and its processes but, if you, as a member of the public or a professional member of the public, see a report that purports to be reporting against the JORC Code and it obviously isn't, the same complaints process applies.

You can lodge that complaint and it'll be investigated. The fact that it's AIM really is irrelevant. We're looking at our own professional standards as opposed to their monitoring and enforcing their particular requirements.

Similarly, if the Competent Person is acting as a qualified person in terms of reporting against National Instrument 43-101 in Canada, complaints would be investigated by the AusIMM again.

So the process is the same wherever you're reporting against the JORC Code.

And of course the Competent Person has to be conscious of the fact that there might be other regulatory requirements there that they may be subject to.

Rene Great. And is there anything else with regards to the application of the JORC Code abroad that you can share with us?

Ian Yes, Rene, look, there are a few areas where I think that there's trends developing and there are areas where I think we have to improve our standards so that the quality of public reporting for the sake of the investor can be improved.

The first is to do with reserves. At the recent CRIRSCO annual meeting, we talked about the consideration of Modifying Factors and how we really have to investigate more those factors applying to the environmental and social aspects of sustainability.

Because problems in those areas can affect the profitability of a venture just as much as the economic and technical ones.

So there really needs to be a better understanding of those things and a better reporting of them, the risks elaborated, because to declare a reserve, there has to be reasonable justification of extraction at the time that you make that declaration of reserve. The other two are more personal.

They're a personal view. One is to do with Competent Persons.

I think we have to lift the bar there. And for AusIMM members, I think that we should be insisting the Competent Person is a chartered professional, a CP.

Part of the CP process is that you have to do continuing professional development, CPD, and that is audited and you have to demonstrate to your peers that you are up to date in your knowledge.

And I would think that one of the CPD areas that a Competent Person should be able to demonstrate is that they have had training in the JORC Code, reporting according to the JORC Code.

And this course that's being developed the certificate course is such a training it's an excellent development in that area. So I would see that as the way to go.

Perhaps it'll be compulsory, perhaps doing the certificate will be compulsory too, but that would do quite a lot to improve the standards of the Competent Person.

We have to keep in mind that the AIG's requirements so that we're, there's a good coordination there and they have their RPGO class of member now so they're thinking that way too I believe.

The other thing is that some of our fellow CRIRSCO members have higher standards. Instead of five years of relevant experience they have seven years, such as Brazil And the SME require that you have three years in a position of seniority in that field.

So that's lifting the bar in terms of experience but also perhaps in management of the process.

So I think that those are are areas that were worth looking at. The second personal point is more emphasis on reasonable prospects of eventual economic extraction.

RPEEE, that's a requirement for the declaration of a resource. And sometimes you get the feeling that not enough effort has been put into that side of the resource estimation.

And perhaps the time will come when a Scoping Study is required before a resource could be declared.

And that leads me into Inferred Resources. One of the qualifications of an Inferred Resource is that more than half would, could be converted to an Indicated Resource based on more exploration.

And I think that's an area too that needs focus so that the Competent Person when he declares an Inferred Resource has confidence that that would be the result. Some of our colleagues think that in this lower end of the spectrum of reporting we need to lift our game a bit.

Rene Thanks, Ian. Really appreciate your time.

Ian Thanks, Rene. I'm happy to contribute. So the Code has a long history and it's widely respected on the international stage.

The Code is now regarded as the world leader in this field with other countries using it as a template for their own reporting standards, in Australia playing a leading role in the development of international reporting standards.

The reasons for the success of the code are varied for most amongst them being its adoption in full by the Australian and New Zealand securities exchanges and the ability and willingness of mining industry professional organisations to bring Competent Persons to account when necessary.

Annual Reporting of Mineral Resources and Ore Reserves

The JORC Code, as well as ASX Listing Rules 5.20 and 5.21, require companies to review and report Mineral Resources and Ore Reserves holdings annually at a date nominated by the company. Companies must publish a resource and reserve statement that includes a discussion on the material changes at the time of publishing, and they must include a comparison with the previous year's statement.

If a company reports its Mineral Resources and Ore Reserves holdings on a date which is not its financial year balance date, a brief explanation of any material change in its holdings between that date and the financial year balance date is required.

The company must also provide a summary of governance arrangements, for example, the reconciliation of Mineral Resource and/or the Ore Reserve.

ASX Listing Rules 5.22 and 5.24 include the regulations Be sure to build your familiarity of both the JORC Code and the ASX Listing Rules as you work through this course. By understanding the nature of these documents, you will build an awareness of when you need to refer to them. You don't need to know them verbatim, but you do need to know what's in them and how to use the search and retrieve information from them.

Interview with Louis Rozman

Rene Sterk interviews Louis Rozman, Co-founder of Pacific Road Capital and former Chair of the VALMIN Committee.

Rene Now to discuss how the JORC Code can help mitigate investment risk, I'm talking with Louis Rozman, who's Operating Partner at Mining Private Equity Firm Pacific Road Capital and also is the Chair of the VALMIN Committee.

Louis, from a market efficiency point of view can you explain why the JORC Code is an important piece of guidance and how it benefits the market?

Louis Thanks Rene, that's a good question. The primary benefit of the JORC Code is it gives investors in our sector a common language so they can all understand what we're talking about.

If we go back prior to the JORC Code, if we go prior to the 1980s, we had companies around the world, bodies around the world using a code in Russia that was different to the code that was being used in China, that was different to the code being used by BHP that was different to the code being used by Newmont.

So you had people all around the world and specifically people in our financial markets in Australia, the USA and Canada, that were that were getting a different story about what a resource and what a reserve was.

And as a result, you had investors who were misinformed whilst investing in the market. And the whole benefit of the Code is that everybody is on a level playing field with a similar understanding of what is intended about the potential of these assets.

Rene Okay. And what happens if people do not adhere to the Code? What are the consequences? Now an often heard argument is that investors get what's coming to them if they choose to invest without doing their own proper due diligence.

We sometimes hear that it is their own risk and that guidance such as the JORC Code only creates this barrage of red tape for companies preventing them from raising funds in a very competitive environment.

Do you think that more regulation is good? And does the JORC Code suit its purpose? Does it indeed contribute towards a more transparent marketplace?

Louis Let's not go down the avenue of is more regulation is good. Let's, let's argue, is there a is there a real benefit for investors in the mining space for having something like the JORC Code?

Investors can invest in a whole lot of things. You know, I remember when I was in the gold industry, you know, we realised that we were competing against companies that sold widgets, we were competing against companies that sold cars, we're competing against people, companies that that sold pharmacy products.

People understand a lot of the market. If you are buying real estate in Sydney there is endless comparables of real estate in Sydney or Kalgoorlie or Melbourne or Perth that allows you to understand what your this property you're considering purchasing is worth in comparison to others that it competes with.

If you're buying classic cars, you know, there is a portfolio of classic cars available for analysis every day of the week and you can determine whether the price for the car you're considering is the right price for the car.

If you're looking at banking stocks, you know, there are comparables between all the banking stocks, and if you want to invest in banking stocks you can see whether the performance of the bank you're considering is equivalent to the others in the market or it might be better value and therefore you want to buy it.

When people look at mining companies, specifically earlier stage companies that have a lot of their value in resources and reserves, they don't understand those resources and reserves unless there is enough regulation around what they mean for them to make a reasonable judgement. And the whole purpose of the JORC Code is effectively to set companies up so they have a balance sheet of value prescribed to those resources and reserves that should be very similar from company to company.

Rene Thanks for taking time out of your busy schedule for us Louis, I really appreciate.

Louis That's fine, Rene.

Interview with James Rowe

Rene Sterk interviews James Rowe, Manager of ASX Listing Compliance and Member of JORC.

Rene Now to discuss how the JORC Code can help mitigate investment risk. I'm talking with James Rowe, who is Manager of ASX Listings Compliance and also sits on the JORC Committee.

James, from a market efficiency point of view can you explain why the JORC Code is an important piece of guidance in, I guess, how it benefits the market?

James Thank you, and thank you for having me. The mining sector in Australia is a very, very important industry in terms of investment, employment, and export.

It's the largest sector on the ASX. To fund investment, resource companies have to compete with all other companies in the capital markets. So communicating with investors is very important for those companies.

And communicating effectively on resources and reserves is integral to accessing those capital markets.

The JORC Code and the Listing Rules play a significant role in providing consistency and confidence to public reporting by listed entities and to their investors.

It's a principles-based approach that promotes the disclosure of key technical assumptions and information that enable investors to make informed decisions. This, in turn, promotes greater efficiencies in the capital markets for the listed entities.

Rene Now, an often heard argument is that investors get what's coming to them if they choose to invest without doing their proper due diligence.

We sometimes hear that it's their own risk and that guidance, such as the JORC Code, only create this barrage of red tape for companies preventing them from raising funds in a very competitive environment.

Do you think that more regulation is good? Does the JORC Code suit its purpose, and does it indeed contribute towards a more transparent marketplace?

James So in answering that question, you need to remember that the JORC Code's been part of the Listing Rules since 1989.

It's been part of our rules for nearly 30 years, and it's very well known to the Australian investing community.
It's had a number of revisions, and in 2012 the JORC Code and the Listing Rules had a comprehensive review and the outcome was an enhanced disclosure framework.

The Listing Rules and the JORC Code balance the adequacy of disclosure to ensure a fair and efficient market against the regulatory burden of compliance.

Together, the two achieve that aim. The Listing Rules establish a framework for the disclosure of Exploration Results, resources, reserves and production targets.

This is in addition to the listing company's obligations under its continuous disclosure regime and its obligations under the Corporations Act, which govern false and misleading statements.

Underpinning the Listing Rules, of course, is the JORC Code and its principles-based approach to disclosure. Those principles being Materiality, Transparency, and Competence.

Competence, of course, being key in that the information's prepared by an expert. So in answer to your question, the JORC Code and the Listing Rules are dynamic, and they'll always evolve to respond to the needs of the capital markets.

Rene Thanks, James. Is there anything else with regards to the JORC Code from an ASX compliance perspective of things that you want to share with us?

James Sure, a Competent Person must not only be an expert in their field, but they've also got to be proficient in their knowledge of the JORC Code. But it doesn't just end there.

There's also the Listing Rules and the Corporations Act obligations which apply, and this is because investors rely heavily on what the Competent Person has to say in the announcement.

Rene Thanks so much for taking time out of your busy schedule, James. I really appreciate it.

James Pleasure